

Safety Procedures — Nexus Semiconductor Fab 3

Hillsboro, OR · Document ID: NXS-SAF-001 · Revision 3 · Effective 01 Feb 2026

Document ID	NXS-SAF-001	Revision	3
Plant	Nexus Semiconductor Fab 3	Effective Date	01 Feb 2026
Document Owner	ESH Director: Y. Park	Classification	Controlled — Site Use

1. Purpose & Scope

This document specifies environmental, safety, and health procedures for Nexus Semiconductor Fab 3 (Hillsboro, OR), a 300 mm logic facility. Scope includes lithography, etch, deposition, implant, CMP, and the sub-fab gas/chemical distribution areas. Conforms with SEMI S2/S8 and Oregon OSHA.

2. Regulatory & Standards Basis

Applicable: SEMI S2/S8, OSHA HCS, EPA Tier II, Cal/OSHA equivalents.

3. Site Hazard Inventory

The principal site hazards at Nexus Semiconductor Fab 3 (Hillsboro, OR) include: pyrophoric/toxic gases (silane, arsine), HF, cleanroom ESD/contamination. The full hazard inventory is summarized below; refer to the linked SOPs for detailed controls.

ID	Hazard	Primary Control
H-01	Pyrophoric gases (silane)	Double-containment lines; auto-shutoff on flow excursion.
H-02	Toxic gases (arsine, phosphine)	Gas cabinet scrubbers; continuous ppb-level monitoring.
H-03	Hydrofluoric acid (HF)	Calcium gluconate kits at every wet bench; medical surveillance for handlers.
H-04	Ionizing radiation (implant)	Tool interlocks; quarterly dosimetry; refer to RAD-NXS-04.
H-05	Cleanroom contamination	ESD garments; bunny-suit gowning per GWN-NXS-001.

4. Procedures

4.1 Gas Cylinder Change

Performed only by certified Gas Specialists, two-person rule. Purge cycles documented automatically by the gas cabinet PLC and uploaded to the EHS data lake. Reference: SOP-NXS-GAS-11.

4.2 HF Exposure Response

Skin contact: rinse 5 min, apply calcium gluconate gel, transport to on-site medical. Inhalation: remove from area, calcium gluconate nebulization per medical protocol MED-NXS-22.

4.3 Chemical Spill in Sub-Fab

Tier 1 (<1 gal): operator clean-up with PPE. Tier 2: facilities response team. Tier 3: external HAZMAT. Tier escalation matrix in ERP-NXS-002.

4.4 Tool Maintenance Permits

Issued through the Fab Tool Permit System. No permit issued unless tool is in 'Down-Safe' state with vacuum broken, chamber purged, and energy isolated.

5. Roles & Responsibilities

The ESH Director owns this procedure. Line Supervisors are accountable for daily compliance. Employees and contractors are responsible for stopping work in the event of an unsafe condition (Stop Work Authority).

6. References

- SEMI S2-0712 — Safety Guidelines for Semiconductor Manufacturing Equipment
- SEMI S8 — Ergonomics
- OSHA 29 CFR 1910.119 — PSM (applicable HPMs)
- NFPA 318 — Cleanrooms
- Internal: GWN-NXS-001, RAD-NXS-04, SOP-NXS-GAS-11, ERP-NXS-002, MED-NXS-22

7. Revision History

Rev	Date	Author	Summary
1	12 Mar 2024	Y. Park	Initial issue.
2	08 Aug 2025	Y. Park	Added confined-space monitoring requirements.
3	01 Feb 2026	Y. Park	Annual review; aligned with updated regulator guidance.

Approvals

Role	Name	Signature	Date
ESH Director	Y. Park	_____	01 Feb 2026
Quality Assurance	V. Acharya, Yield & Quality Manager	_____	01 Feb 2026
Operations / Site Lead	Q. Sandberg, Fab GM	_____	01 Feb 2026

CONFIDENTIAL — Controlled Document. Uncontrolled when printed. Reproduction outside the parent organization requires written authorization.